FollowUp AI

Patient Follow-up Risk Predictor — System Architecture & Workflow

Purpose: Predict which patients are most likely to miss their next follow-up appointment, explain the risk, rank patients for staff attention, and track interventions.

1. HIGH-LEVEL ARCHITECTURE

The application is designed as a modular healthcare operations platform. The current prototype can use local persistence and a transparent scoring engine; the scoring layer can later be replaced by a trained ML classifier without redesigning the user interface.

Patient Data	↓	Risk Engine	↓	Explainability	↓	Priority Queue	↓	Staff Action	↓	Outcome Tracking

Architecture layers

Presentation Layer	Dashboard • Patients • Priority Queue • Appointments • Analytics • Risk Simulator • Settings
Application Layer	Patient management • Search/filter/sort • Risk calculation • Recommendations • Intervention workflow • Notifications
Risk / AI Layer	Weighted scoring model today; replaceable with historical-data ML classifier later. Returns score, category, factor contributions, explanation, and recommendation.
Data Layer	Patient records • Appointment history • Intervention records • Model weights • Demo dataset • LocalStorage/database
Integration Layer	Future: EHR/EMR • SMS/WhatsApp • Hospital scheduling systems • Authentication • Analytics/monitoring

2. RISK PREDICTION ENGINE

The prototype uses a transparent 0–100 weighted risk model. Each factor produces a visible contribution so staff can understand why a patient is prioritized.

Factor	Maximum Points	Example Logic
Missed appointments	40	0 → 0; 1 → 10; 2 → 20; 3 → 30; 4+ → 40
Attendance history	15	Higher missed/low attendance rate increases contribution
Distance	15	<5 km → 0; 5–10 → 5; 10–20 → 10; >20 → 15
Treatment duration	10	Longer duration contributes more
Appointment frequency	10	More frequent appointments contribute more
Age	10	Minor contribution; capped by predefined bands

Risk categories: 0–29 = Low • 30–59 = Medium • 60–100 = High

Engine output: score + category + factor contributions + plain-language explanations + recommended action.

3. END-TO-END WORKFLOW

Enter / Import Patient Data	↓	Validate Data	↓	Calculate Risk	↓	Generate Reasons	↓	Rank Patients	↓	Recommend Action	↓	Track Outcome

4. STAFF WORKFLOW

Step	What happens	Staff result
1. Identify	Dashboard summarizes high-, medium-, and low-risk patients.	See who needs attention.
2. Prioritize	Priority Queue sorts patients by risk and appointment urgency.	Work highest-risk patients first.
3. Understand	Patient detail shows factor contributions and appointment history.	Know why the patient is risky.
4. Intervene	Staff can call, send a reminder/SMS, reschedule, or mark contacted.	Take a preventive action.
5. Track	Record intervention status and notes.	Maintain an auditable workflow.
6. Learn	Analytics compares risk distribution and intervention outcomes.	Improve operational decisions.

5. DETAILED SYSTEM FLOW

Patient record enters the system → validation checks required fields and acceptable ranges → scoring engine calculates each factor → explanations are generated from the same factor values → patients are ranked → staff receive an intervention recommendation → intervention is recorded → outcome feeds analytics and future model development.

INPUT	PROCESS	DECISION	ACTION	FEEDBACK
Age Missed visits History Distance Treatment Frequency Next appointment	Validation Feature calculation Weighted scoring	0–29 30–59 60–100	Call SMS Reminder Reschedule Mark contacted	Attended Missed Rescheduled Unable to reach

6. CORE APPLICATION MODULES

Module	Function
Dashboard	Hospital-level risk summary, upcoming follow-ups, intervention alerts
Patients	Search, add, edit, delete, and inspect patient records
Priority Queue	Risk-ranked worklist with filters and intervention status
Appointments	Upcoming visits and historical attendance timeline
Interventions	Call/SMS/reminder/reschedule actions and notes
Analytics	Risk distribution, missed-visit trends, and intervention outcomes
Risk Simulator	Change factors and observe score changes in real time
Settings	Risk weights and model transparency

7. FUTURE ML ARCHITECTURE

Historical Attendance Data	↓	Feature Engineering	↓	ML Classifier	↓	Probability Calibration	↓	Explainability Layer	↓	Risk Queue

The transparent rules-based prototype is intentionally replaceable. With sufficient historical attendance data, a supervised classifier can estimate the probability of missing the next follow-up. The production version should validate calibration, monitor model performance, and evaluate fairness before clinical/operational deployment.

8. SAFETY & GOVERNANCE

This system is decision support for appointment follow-up operations, not diagnosis or treatment. Predictions should be reviewed by authorized staff. Production deployment should include access control, audit logs, data minimization, privacy/security controls, model monitoring, and validation on representative local data.

Core Product Loop

PREDICT → EXPLAIN → PRIORITIZE → INTERVENE → TRACK → IMPROVE